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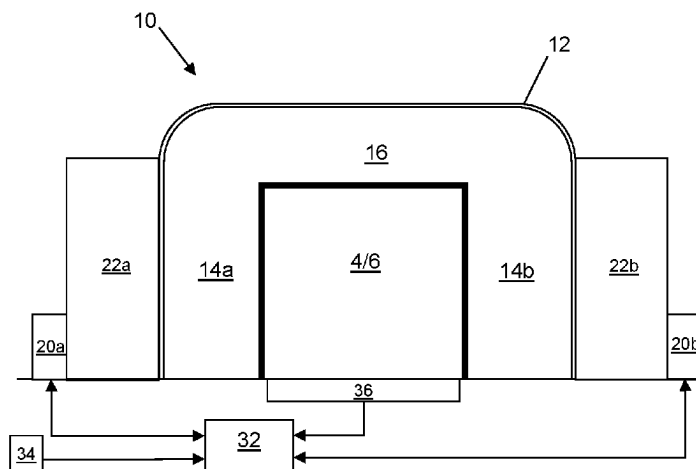
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(52) **U.S. Cl.**
CPC *F24F 9/00* (2013.01); *F24F 11/30*
(2018.01); *F24F 2009/005* (2013.01);
(Continued)

(57) **ABSTRACT**

An active airflow inhibiting apparatus for an entranceway comprises: a structure positioned adjacent an entranceway of a building and defines a passage therethrough for accessing the entranceway. The structure defines at least one plenum chamber and first and second outlet slots fluidically coupled to the plenum chamber. A fan fluidically connected to the plenum chamber supplies an airflow to the plenum chamber in order to selectively form a jet of air from the first outlet slot in a first mode of operation and from the second outlet slot in a second mode of operation. The first and second outlet slots are configured so that the respective jets of air are each directed towards a centre of the structure, with the jet of air from the first outlet slot being directed away from the

(Continued)



entranceway and the jet of air from the second outlet slot being directed towards the entranceway.

20 Claims, 4 Drawing Sheets

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(58) **Field of Classification Search**

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See application file for complete search history.

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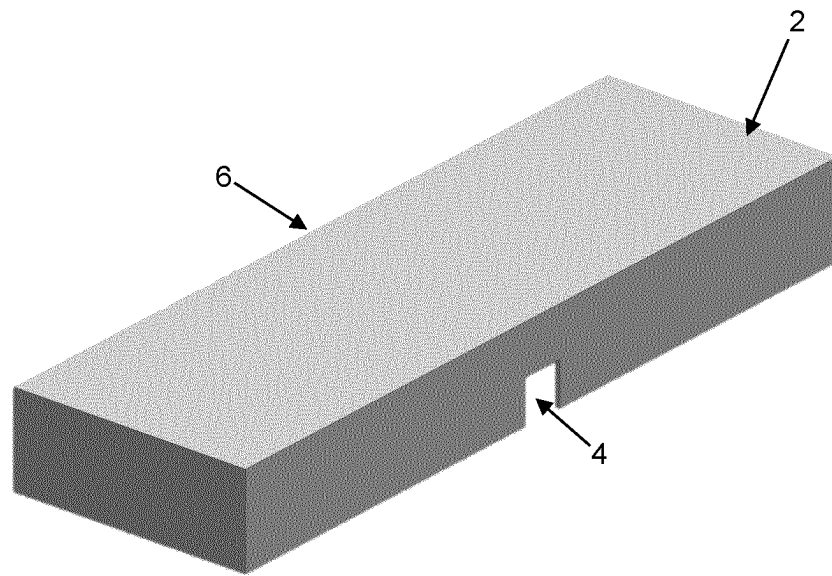


FIG. 1

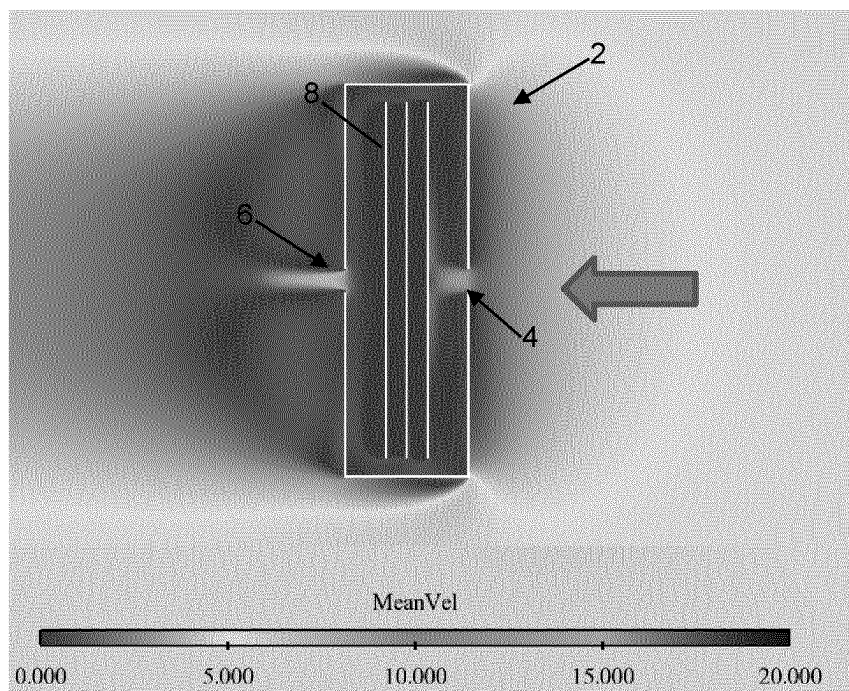


FIG. 2

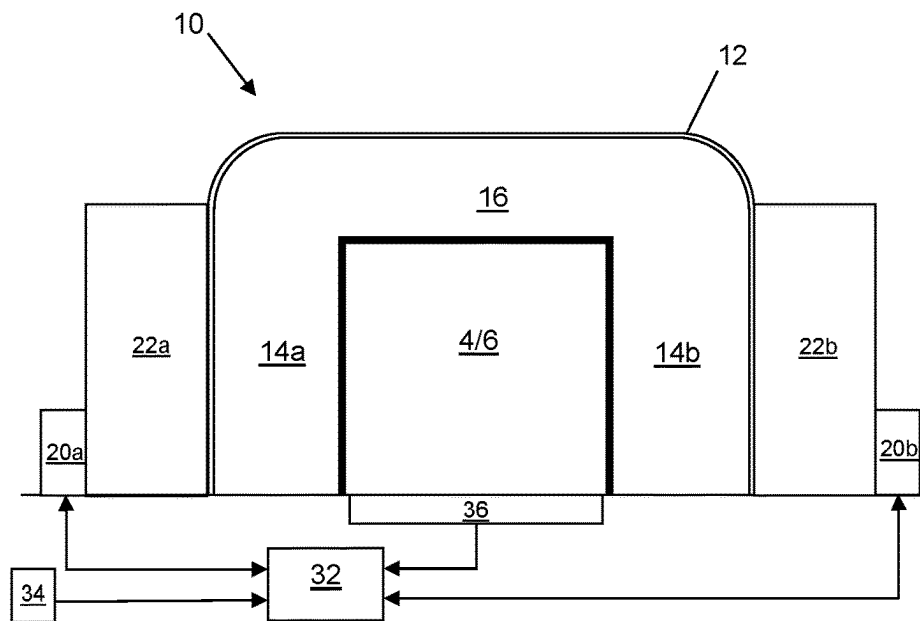


FIG. 3

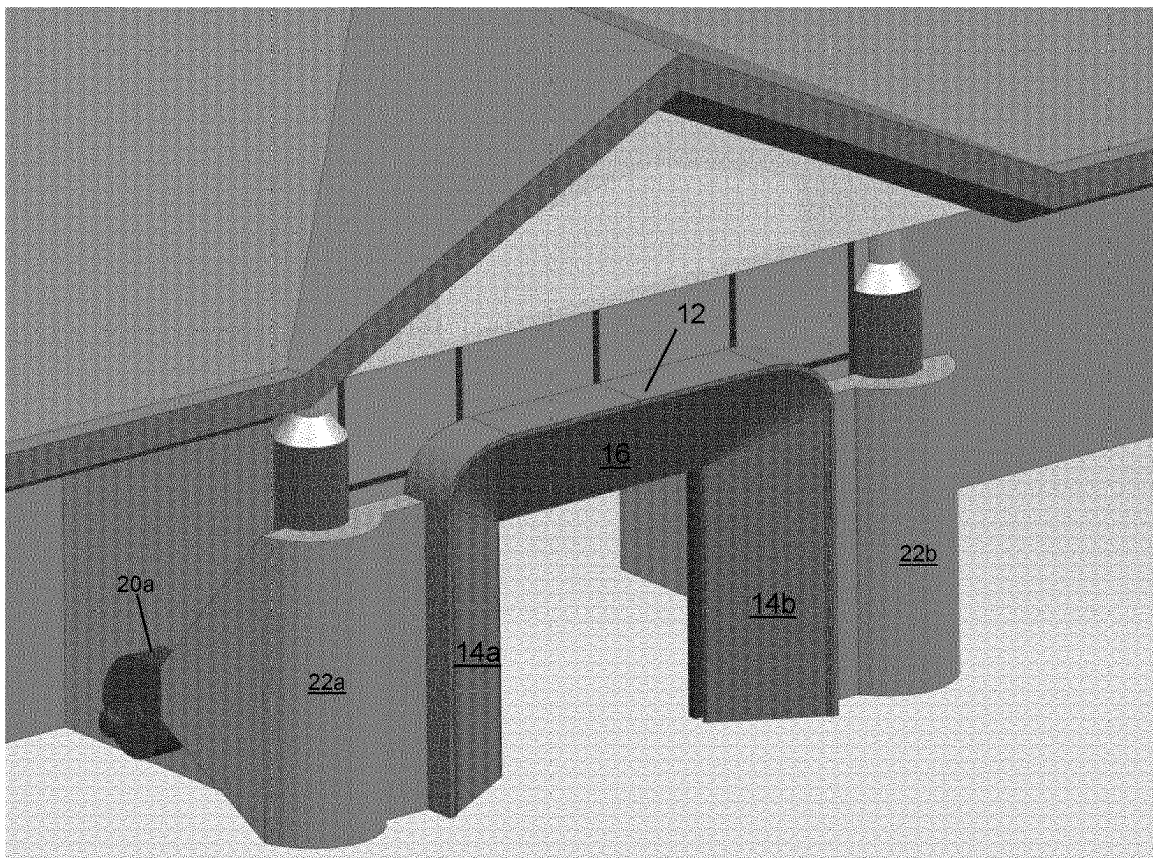


FIG. 4

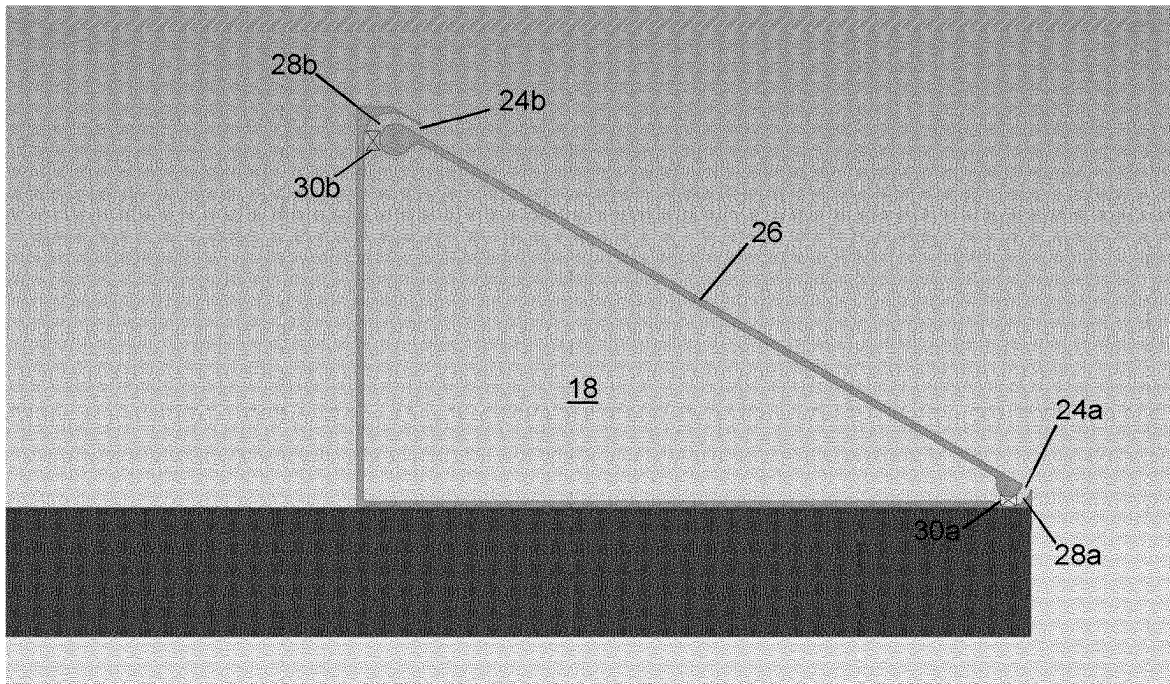


FIG. 5

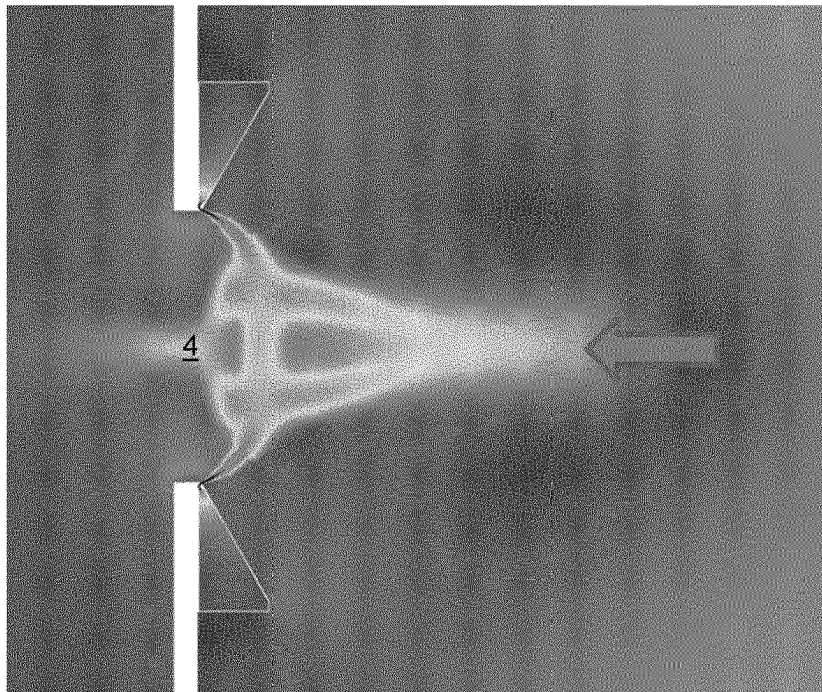


FIG. 6

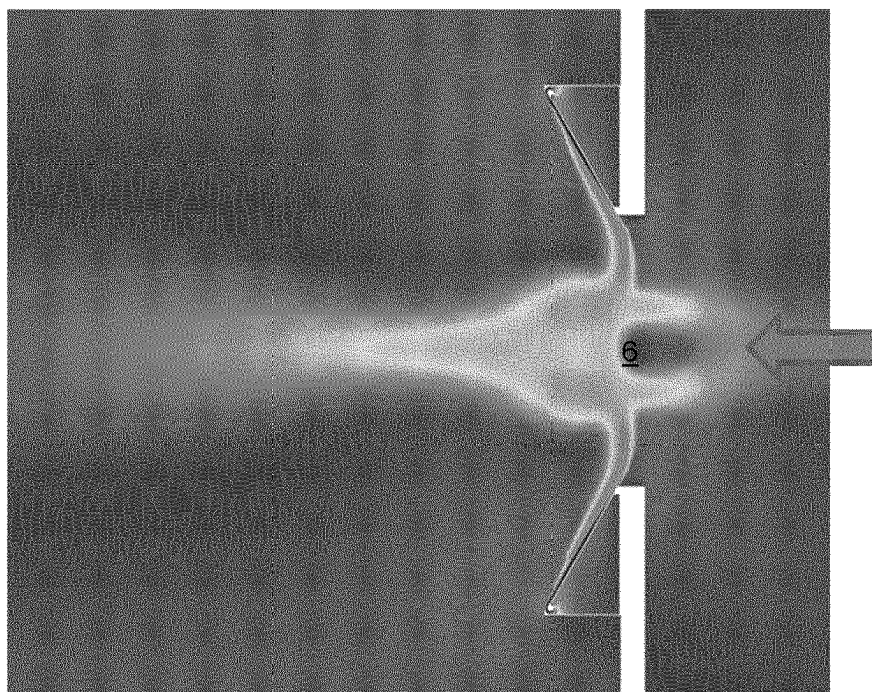


FIG. 7

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**ACTIVE AIRFLOW INHIBITING
APPARATUS****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is the U.S. national stage of PCT/EP2020/062037 filed on Apr. 30, 2020, which claims priority of Great Britain Application No. GB1906016.9 filed on Apr. 30, 2019, the contents of which are incorporated herein.

TECHNICAL FIELD

The disclosure relates to an active airflow inhibiting apparatus for an entranceway, and particularly a doorway.

BACKGROUND

Various styles of doors are used at the entrances of retail spaces, such as shopping centres (malls), supermarkets or other stores.

A common style of door is the automatic sliding door. Two sets of automatic sliding doors are often provided in series to form a draught lobby which acts as an airlock to prevent wind from entering the building. However, in high traffic areas, it is common for both sets of doors to be open at the same time, thus providing a direct path for air to pass through the door. This can lead to unpleasant draughts. In addition, airflow may occur through a door as a result of temperature differentials across the door. Airflow through a door (whether from draughts or induced by temperature differentials) increases the power output requirements of HVAC systems within the building.

Over door heaters are often used to try to mask the incoming draught to improve customer experience. However, these devices consume large amounts of energy and do not address the problem itself. Another option is to provide an air curtain across the doorway. However, these devices are not able to prevent infiltration where there is a large pressure differential or under windy conditions.

Experience has shown that any physical barrier, even an automatically opening door, can lead to a reduction in the number of people entering a store and that stores have started keeping the doors open throughout all of the working hours to minimise this effect. In this case, the energy costs through the passage of heated or cooled air out of the building and replaced by ambient air can be substantial.

It is therefore desirable to provide an airflow inhibiting apparatus which addresses the disadvantages of existing solutions.

SUMMARY

In accordance with an aspect of the disclosure, there is provided an active airflow inhibiting apparatus for an entranceway of a building comprising: a structure configured to be positioned adjacent the entranceway and defining a passage therethrough for accessing the entranceway; wherein the structure defines at least one plenum chamber and first and second outlet slots fluidically coupled to the at least one plenum chamber; a fan fluidically connected to the plenum chamber for supplying an airflow to the at least one plenum chamber in order to selectively form a jet of air from the first outlet slot in a first mode of operation and from the second outlet slot in a second mode of operation; wherein the first and second outlet slots are configured so that the respective jets of air are each directed towards a centre of the

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structure, with the jet of air from the first outlet slot being directed away from the entranceway and the jet of air from the second outlet slot being directed towards the entranceway.

5 The at least one plenum chamber may be fluidically coupled to the first outlet slot via a first curved passageway and fluidically coupled to the second outlet slot via a second curved passageway.

10 A valve may be disposed between the at least one plenum chamber and the first and second outlet slots, the valve being selectively controllable to change between the first and second modes of operation.

15 The structure may comprise a Coandă surface and the first and second outlet slots may be spaced from one another along the Coandă surface.

The Coandă surface may be inclined with respect to the plane of the entrance.

20 The Coandă surface may be configured to guide the jet of air from the second outlet slot along its length past the first outlet slot and towards the entranceway.

The first outlet slot may extend around an inner perimeter of the structure and the second outlet slot may extend around an outer perimeter of the structure.

25 The structure may have a triangular cross-section which tapers towards its inner perimeter.

The structure may form an archway.

30 The active airflow inhibiting apparatus may further comprise: a controller configured to control the jet of air from the first and second outlet slots to provide a differential pressure across the structure which inhibits airflow through the entranceway.

35 The apparatus may comprise a plurality of said structures and each of the plurality of said structures may be configured to be positioned adjacent a different entranceway of the same building.

40 The controller may be configured to determine a set of operating parameters (e.g. fan speed setting, valve position, etc.) for the plurality of structures which are dependent on one another.

45 The active airflow inhibiting apparatus may further comprise: an airflow sensor configured to provide an output indicative of speed and direction of airflow through the entranceway or at the structure; and the controller may be configured to receive the output of the airflow sensor and to control the jet of air based on the received output so as to generate a differential pressure across the structure which inhibits airflow through the entranceway.

50 The airflow sensor may be configured to provide an output indicative of speed and direction of airflow through the entranceway at a plurality of vertical positions through the entranceway.

55 The airflow sensor may comprise a plurality of sensor elements located at different vertical positions.

The controller may be configured to control the jet of air so as to generate a differential pressure which varies with vertical position.

60 The controller may be configured to synchronise the operation of the fans with the opening of a door of the entranceway based on the output of an activation sensor.

The activation sensor may be located within the passage defined by the structure.

65 The jet of air may be controlled by changing a fan speed setting.

The structure may be configured to be located externally to the entranceway.

BRIEF DESCRIPTION OF DRAWINGS

For a better understanding of the disclosure, and to show more clearly how it may be carried into effect, reference will now be made, by way of example, to the accompanying drawings, in which:

FIG. 1 is a model of a store building;

FIG. 2 is an airflow velocity plot over a plan view of the store building showing airflow around and through the building;

FIG. 3 is a front view of an airflow inhibiting apparatus according to an embodiment of the disclosure;

FIG. 4 is a perspective view of the airflow inhibiting apparatus;

FIG. 5 is a cross-section through a portion of the airflow inhibiting apparatus;

FIG. 6 is an airflow velocity plot over a plan view of the building showing airflow adjacent the front entrance with the airflow inhibiting apparatus in use; and

FIG. 7 is an airflow velocity plot over a plan view of the building showing airflow adjacent the rear entrance with the airflow inhibiting apparatus in use.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

FIGS. 1 and 2 show a simplified model of a store building 2, such as a supermarket or other retail space. As shown, in this example, the building 2 has a front entrance 4 and a rear entrance 6 located opposite the front entrance 4. The front and rear entrances 4, 6 are shown as open doorways to signify that a door located at the entrance is in an open position and so not covering the doorway. In particular, the front and rear entrances 4, 6 may utilise automatic sliding doors and so the model shown in FIG. 1 represents where customers are passing simultaneously through the front and rear entrances 4, 6.

As shown in FIG. 2, internally, the space within the building 2 is divided into a number of aisles by dividers 8.

With wind directed towards the front entrance 4 as shown in FIG. 2, air is allowed to flow into the building 2 via the front entrance 4 and is drawn through the building 2 before exiting the building 2 via the rear entrance 6. A strong current of air (draught) is therefore generated through the interior of the building 2. This may be unpleasant for customers and employees located within the building 2.

FIGS. 3 and 4 show an airflow inhibiting apparatus 10 which may be provided adjacent the front entrance 4 and/or rear entrance 6 and seeks to reduce or eradicate entirely such draughts through the building.

As shown in FIG. 3, the airflow inhibiting apparatus 10 comprises a structure 12 which is positioned adjacent the entrance 4, 6 on the exterior of the building 2. In particular, the structure 12 may be disposed against an external wall of the building 2. The structure 12 forms an arch which extends around the entrance 4, 6 such that it does not impede the doorway. Specifically, the structure 12 comprises first and second vertical sections 14a, 14b which extend along either side of the entrance 4, 6 and a horizontal section 16 which is disposed above the entrance 4, 6. The first and second vertical sections 14a, 14b transition into the horizontal section 16 at first and second corner sections.

As shown in FIG. 5, the structure 12 is hollow and forms a plenum chamber 18. The plenum chamber 18 may be continuous or may be divided into a plurality of discrete

sections. For example, the plenum chamber 18 may be divided into two discrete sections along the axis of symmetry of the structure 12.

As shown in FIGS. 3 and 4, the structure 12 is fluidically connected to a pair of fans 20a, 20b via first and second ducts 22a, 22b. Specifically, the first duct 22a connects the fan 20a to the first vertical section 14a of the structure 12 and the second duct 22b connects the fan 20b to the second vertical section 14b of the structure 12.

The structure 12 defines a first outlet slot 24a and a second outlet slot 24b which are fluidically coupled to the plenum chamber 18. As shown, the first outlet slot 24a extends around an inner perimeter of the structure 12 at or adjacent the perimeter of the entrance and the second outlet slot 24b extends around an outer perimeter of the structure 12. The first outlet slot 24a is thus nested within the second outlet slot 24b (i.e. they are concentric or coaxial) and the second outlet slot 24b is spaced from the first outlet slot 24a and the perimeter of the entrance.

The second outlet slot 24b is spaced from the first outlet slot 24a along a Coandă surface 26. The Coandă surface 26 is inclined with respect to the plane of the entrance (i.e. neither parallel nor perpendicular to the plane of the entrance) such that the second outlet slot 24b is spaced further from the plane of the entrance than the first outlet slot 24a. The structure 12 therefore has a generally triangular (specifically, a right-angled triangular) cross-section and is thus wedge-shaped (specifically, a right triangular prism) tapering towards its inner perimeter adjacent the entrance.

The first outlet slot 24a is coupled to the plenum chamber 18 via a first curved passageway 28a and the second outlet slot 24b is coupled to the plenum chamber 18 via a second curved passageway 28b.

A first valve 30a is provided between the first outlet slot 24a and plenum chamber 18 (e.g. in the first curved passageway 28a) and a second valve 30b is provided between the second outlet slot 24b and plenum chamber (e.g. in the second curved passageway 28b). The first and second valves 30a, 30b may be gate valves. The first and second valves 30a, 30b act in unison to selectively open one of the first and second outlet slots 24a, 24b at a time, as will be described further below. It will be appreciated that in other examples, a single valve may be able to simultaneously open one of the first and second outlet slots 24a, 24b and close the other of the first and second outlet slots 24a, 24b.

The fans 20a, 20b provide an airflow to the plenum chamber 18 to form a pressurised volume within the structure 12. The air is released from the plenum chamber 18 via one of the first and second outlet slots 24a, 24b, forming a jet of air.

The first and second curved passageways 28a, 28b act to direct the jet of air in the desired direction. Specifically, the first and second curved passageways 28a, 28b both act to direct the respective jets of air from the first and outlet slots 24a, 24b so that they are directed towards the centre of the structure 12. In other words, the air from the first vertical section 14a is directed towards the opposing second vertical section 14b, and vice versa, and the air from the vertical section is directed downwards towards the ground. However, the first curved passageway 28a acts to direct the jet of air from the first outlet slot 24a so that it is directed away from the entrance and the interior of the building 2, whereas the second curved passageway 28b acts to direct the jet of air from the second outlet slot 24b so that it is directed towards the entrance and the interior of the building 2. Specifically, the second curved passageway 28b acts to direct the jet of air from the second outlet slot 24b so that it flows along the

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Coandă surface **26** towards the first outlet slot **24a**. The Coandă surface **26** causes the jet of air to form a laminar boundary layer along its length.

The fans **20a**, **20b** and the structure **12** form an air mover device. Specifically, the air mover device is a bidirectional air multiplier having a first mode of operation in which a jet of air is ejected from the first outlet slot **24a** and a second mode of operation in which a jet of air is ejected from the second outlet slot **24b**. In either mode, the jet of air creates an area of negative pressure which draws additional air into the airflow from around the structure **12**. Further, as the air moves away from the structure **12** it entrains additional air within the airflow. The volume of air within the airflow is thus multiplied. The selection of the first and second modes of operation may be controlled by an internal controller of the air mover device which operates the valves **30a**, **30b**.

The air mover device **12** is connected (either via a wired or wireless connection) to a controller **32** which is in turn connected (again, either via a wired or wireless connection) to an airflow sensor **34** and an activation sensor **36**.

The activation sensor **36** may be a pressure sensor or a movement sensor (such as a passive infra-red sensor or the like) which provides a signal that indicates when someone passes through the structure **12** prior to entering the building **2** via the entrance **4**, **6**.

The airflow sensor **34** provides an output which is indicative of the present wind conditions, particularly the wind speed and direction.

The controller **32** receives as inputs the signals from the activation sensor **36** and the airflow sensor **34**. The controller **32** uses these signals to control the operation of the air mover device. Specifically, the controller **32** sets a fan speed setting of the fans **20a**, **20b** based on the speed and direction of the wind. The fan speed setting is set to create a pressure differential which opposes the approaching wind and is sufficient to cause it to be substantially stopped, redirected or reversed.

FIG. **6** shows the airflow velocity at the front entrance **4** where the wind enters the building **2**, and FIG. **7** shows the airflow velocity at the rear entrance **6** where the wind exits the building **2**. The air mover device at the front entrance **4** is configured to operate in the first mode of operation where a jet of air is ejected from the first outlet slot **24a** and the air mover device at the rear entrance **6** is configured to operate in the second mode of operation in which a jet of air is ejected from the second outlet slot **24b**. As shown, at both locations, the wind is prevented from passing through the structure **12** and thus the adjacent doorway, thus creating stagnant conditions within the building **2**. The fan speed setting can be controlled based on the speed and direction of the wind to ensure that the jet of air has sufficient power to prevent the wind from passing through the structure **12**.

The operation of the air mover device is also coordinated based on the signals of the activation sensor **36**. Specifically, the fans **20a**, **20b** may only be switched on or operated at the required fan speed (differential pressure) setting when someone is approaching the front entrance and the door will open allowing a draft to be formed. A corresponding sensor may be provided inside the building **2** to indicate when the door will be triggered by someone leaving the building **2**.

If the wind direction were reversed such that air entered the rear entrance **6** of the building and exited the front entrance **4**, the air mover devices would be operated in the opposite configurations.

The controller **32** is able to actively manage the operation of the air mover devices to prevent or minimise draughts at all times, regardless of the current wind conditions. The

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controller **18** may access a look-up table or other reference source to determine the correct setting for the current wind conditions.

The controller **32** may be in communication with each of the air mover devices at the front and rear entrances **4**, **6** and thus be able to make local adjustments to prevent airflow either into or out of the respective doorways. The effect of each of the arrays has an impact on the other arrays and so the settings for the arrays cannot be determined in isolation. Consequently, the controller **32** determines a set of outputs for the air mover devices which are dependent on one another. In particular, the controller **32** may perform a multivariate analysis (or other analysis) which seeks to define the optimum overall solution (particularly, with the minimum energy usage).

Although the airflow inhibiting apparatus **10** has been described in relation to airflows generated by wind, it will be appreciated that it may also minimise or prevent airflows associated with temperature differentials at a doorway (i.e. in the absence of any wind or draught). Such temperature differentials lead to both ingress and egress at the doorway as a result of buoyancy effects. Specifically, higher density, colder air flows in one direction at the lower part of the door plane and lower density, warmer air flows in the opposite direction at the upper part of the door plane in order to maintain net building pressure.

In such circumstances, the airflow sensor **34** is able to determine the current airflow through the doorway at a plurality of vertical positions (for example, by utilising a plurality of sensor elements located at different vertical positions). The controller **32** is able to utilise the output of the airflow sensor **34** to control the output of the air mover device to vary with vertical position. Specifically, the air mover device is able to generate a stratified differential pressure which provides a negative pressure over part of the doorway and a positive pressure over another part of the same doorway in order to counteract the opposing flows through the doorway generated by buoyancy effects.

The airflow may also vary vertically and/or horizontally, while generating a positive or negative pressure all around the structure **12**, to take into account variations in wind conditions and directions.

The front of the building may comprise a recess (for example, being dished inwardly), with the doorway being positioned within the recess so that it is set back from the boundary of the building. This arrangement may allow the structure **12** to be sited within or at the boundary of the building (although still external to the doorway).

The preceding description describes how the output of the air mover devices is controlled by adjusting a fan speed setting. In other arrangements, the output of the air mover devices may be adjusted in other manners. For example, the output may be adjusted by controlling valves/chokes (such as the valves **30a**, **30b**) or by adjusting the size of the outlet slots **24a**, **24b**.

Although the airflow sensor **34** is shown as being adjacent to the structure **12**, it will be appreciated that the airflow sensor may be located remotely provided that it gives an adequate indication of the current wind conditions at that location.

The activation sensor **36** may be omitted in other examples or may be formed by the opening sensor of the door itself.

In other examples, the structure **12** may not form an arch. For example, the structure may comprise a pair of vertical sections (with a passage therebetween), and optionally, a horizontal section. Further, it is not necessary for the entire

arch to generate airflow. For example, the first and second slots **24a**, **24b** may not extend over the corners of the arch.

The airflow may be provided by any number of fans. The or each fan may also connect directly to the plenum chamber rather than via an intermediate duct.

Although it has been described that the structure **12** has first and second outlet slots **24a**, **24b**, it will be appreciated that the first and second outlet slots **24a**, **24b** may be divided into a plurality of discrete sections.

The airflow inhibiting apparatus **10** may only be provided on a single entrance of a building. In particular, this may be sufficient to prevent airflow through the building even when there are other entrances.

The airflow inhibiting apparatus **10** is able to inhibit airflow (generated by wind and/or temperature differentials) through a doorway (or any other entranceway) without requiring any physical obstruction. This improves customer experience and reduces power consumption of HVAC systems operating within the building.

The disclosure is not limited to the embodiments described herein, and may be modified or adapted without departing from the scope of the present disclosure, as defined by the claims.

The invention claimed is:

1. An active airflow inhibiting apparatus for an entranceway of a building comprising:

a structure configured to be positioned adjacent the entranceway and defining a passage therethrough for accessing the entranceway;

wherein the structure defines at least one plenum chamber and first and second outlet slots fluidically coupled to the at least one plenum chamber;

a fan fluidically connected to the at least one plenum chamber for supplying an airflow to the at least one plenum chamber in order to selectively form a jet of air from the first outlet slot and not the second outlet slot in a first mode of operation and from the second outlet slot and not the first outlet slot in a second mode of operation;

wherein the first and second outlet slots are configured so that the respective jets of air are each directed towards a center of the structure, with the jet of air from the first outlet slot being directed away from the entranceway and the jet of air from the second outlet slot being directed towards the entranceway.

2. The active airflow inhibiting apparatus as claimed in claim 1, wherein the at least one plenum chamber is fluidically coupled to the first outlet slot via a first curved passageway and is fluidically coupled to the second outlet slot via a second curved passageway.

3. The active airflow inhibiting apparatus as claimed in claim 1, wherein a valve is disposed between the at least one plenum chamber and the first and second outlet slots, the valve being selectively controllable to change between the first and second modes of operation.

4. The active airflow inhibiting apparatus as claimed in claim 1, wherein the structure comprises a Coandă surface and the first and second outlet slots are spaced from one another along the Coandă surface.

5. The active airflow inhibiting apparatus as claimed in claim 4, wherein the Coandă surface is inclined with respect to the plane of the entrance.

6. The active airflow inhibiting apparatus as claimed in claim 4, wherein the Coandă surface is configured to guide

the jet of air from the second outlet slot along its length past the first outlet slot and towards the entranceway.

7. The active airflow inhibiting apparatus as claimed in claim 1, wherein the first outlet slot extends around an inner perimeter of the structure and the second outlet slot extends around an outer perimeter of the structure.

8. The active airflow inhibiting apparatus as claimed in claim 1, wherein the structure has a triangular cross-section which tapers towards its inner perimeter.

9. The active airflow inhibiting apparatus as claimed in claim 1, wherein the structure forms an archway.

10. The active airflow inhibiting apparatus as claimed in claim 1, further comprising:

a controller configured to control the jet of air from the first and second outlet slots to provide a differential pressure across the structure which inhibits airflow through the entranceway.

11. The active airflow inhibiting apparatus as claimed in claim 10, wherein the apparatus comprises a plurality of said structures and wherein each of the plurality of said structures is configured to be positioned adjacent a different entranceway of the same building.

12. The active airflow inhibiting apparatus as claimed in claim 11, wherein the controller is configured to determine a set of operating parameters for the plurality of structures which are dependent on one another.

13. The active airflow inhibiting apparatus as claimed in claim 10, further comprising:

an airflow sensor configured to provide an output indicative of speed and direction of airflow through the entranceway or at the structure; and

wherein the controller is configured to receive the output of the airflow sensor and to control the jet of air from the first and second outlet slots based on the received output so as to generate the differential pressure across the structure which inhibits the airflow through the entranceway.

14. The active airflow inhibiting apparatus as claimed in claim 13, wherein the airflow sensor is configured to provide the output indicative of the speed and direction of the airflow through the entranceway at a plurality of vertical positions through the entranceway.

15. The active airflow inhibiting apparatus as claimed in claim 14, wherein the controller is configured to control the jet of air from the first and second outlet slots so as to generate a differential pressure which varies with vertical position.

16. The active airflow inhibiting apparatus as claimed in claim 13, wherein the airflow sensor comprises a plurality of sensor elements located at different vertical positions.

17. The active airflow inhibiting apparatus as claimed in claim 10, wherein the controller is configured to synchronise the operation of the fan with the opening of a door of the entranceway based on an output of an activation sensor.

18. The active airflow inhibiting apparatus as claimed in claim 17, wherein the activation sensor is located within the passage defined by the structure.

19. The active airflow inhibiting apparatus as claimed in claim 10, wherein the jet of air from the first and second outlet slots is controlled by changing a fan speed setting.

20. The active airflow inhibiting apparatus as claimed in claim 1, wherein the structure is configured to be located externally to the entranceway.